Foundations of Programming

Assignment No.: 18

PROBLEM STATEMENT: Write a C program that accepts a string from the user and performs the following string operations: i) Calculate length of string ii) String reversal iii) Equality check of two strings iv) Check palindrome v) Check substring.

OBJECTIVE:

- To understand basic string operations in C.
- To learn how to manipulate strings using loops and functions.
- To perform comparison, reversal, and substring checking.
- To check whether a string is a palindrome.

THEORY:

In C programming, a string is a sequence of characters terminated by a null character ('\0'). Strings are commonly stored in character arrays. Various operations can be performed on strings such as calculating their length, reversing the characters, comparing two strings, checking whether a string is a palindrome, and finding a substring within another string. These operations can be implemented using loops and conditional statements or by using standard library functions from the string.h header file.

PLATFORM: Linux

ALGORITHM:

1. Start
2. Input a string str1 from the user
3. Find length of string:
- Set length = 0
- Repeat until end of string:
  - Increment length
- Display length
4. Reverse the string:
- For i from last index to first:
  - Print each character
5. Check equality of two strings:
- Input another string str2
- Compare both strings character by character
- If all characters are same → Print “Strings are Equal”
- Else → Print “Strings are Not Equal”
6. Check palindrome:
- Compare characters from start and end
- If all match → Print “Palindrome”
- Else → Print “Not Palindrome”
7. Check substring:
- Input a substring
- Check if it exists inside main string
- If found → Print “Substring Found”
- Else → Print “Substring Not Found”
8. Stop

CODE:

```
#include <stdio.h>

#include <string.h>

int main() {

    char str1[100], str2[100], rev[100];

    int n, i, j;

    printf("Enter string 1: ");

    scanf("%s", str1);

    printf("Enter string 2: ");

    scanf("%s", str2);

    n = strlen(str1);

    printf("\nLength of str1: %d\n", n);

    j = 0;

    for (i = n - 1; i >= 0; i--) {

        rev[j++] = str1[i];

    }

    rev[j] = '\0'; // Null-terminate the string

    printf("Reversed str1: %s\n", rev);

    if (strcmp(str1, str2) == 0) {

        printf("Strings are equal\n");

    } else {

        printf("Strings are not equal\n");

    }

    if (strstr(str1, str2) != NULL) {

        printf("Substring found\n");

    } else {

        printf("Substring not found\n");

    }

    int isPalindrome = 1;

    for (i = 0; i < n / 2; i++) {
```

```
        if (str1[i] != str1[n - i - 1]) {
            isPalindrome = 0;
            break;
        }
    }
    if (isPalindrome) {
        printf("String is palindrome\n");
    }
   else {
        printf("String is not palindrome\n");
    }
    return 0;
}
```

```c
#include <stdio.h>
#include <string.h>

int main() {
    char str1[100], str2[100], rev[100];
    int n, i, j;

    printf("Enter string 1: ");
    scanf("%s", str1);
    printf("Enter string 2: ");
    scanf("%s", str2);

    n = strlen(str1);
    printf("\nLength of str1: %d\n", n);

    j = 0;
    for (i = n - 1; i >= 0; i--) {
        rev[j++] = str1[i];
    }
    rev[j] = '\0'; // Null-terminate the string
    printf("Reversed str1: %s\n", rev);

    if (strcmp(str1, str2) == 0) {
        printf("Strings are equal\n");
    } else {
        printf("Strings are not equal\n");
    }

    if (strstr(str1, str2) != NULL) {
        printf("Substring found\n");
    } else {
        printf("Substring not found\n");
    }

    int isPalindrome = 1;
    for (i = 0; i < n / 2; i++) {
        if (str1[i] != str1[n - i - 1]) {
            isPalindrome = 0;
            break;
        }
    }

    if (isPalindrome) {
        printf("String is palindrome\n");
    } else {
        printf("String is not palindrome\n");
    }

    return 0;
}
```

OUTPUT:

Enter a string: madam

Enter another string for equality check: mad

Length of string = 5

Reversed string = madam

The string is a Palindrome

Strings are not equal

Substring found in main string

SAMPLE INPUT:

Enter a string: madam

Enter another string for equality check: mad

SAMPLE OUTPUT:

Length of string = 5

Reversed string = madam

The string is a Palindrome

Strings are not equal

Substring found in main string

CONCLUSION: Thus, the C program successfully performs different string operations such as length calculation, reversal, equality check, palindrome check, and substring detection.

FAQs:

1. What is a string in C programming?
2. Which header file is required for string functions?
3. What function is used to find the length of a string?
4. What is a palindrome string?
5. Which function is used to check a substring in C?